

7.4.2 Method of Maximum Likelihood

An unbiased estimator is often determined using trial and error. The method of maximum likelihood, however, is characterized by a step-by-step procedure. It is a commonly used method. An estimator determined in this way is called a maximum likelihood estimator (MLE).

The method is as follows:

- Define the likelihood function $L(\theta)$:

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \dots \cdot f(x_n; \theta).$$

- Determine the value of θ that maximizes $L(\theta)$.
- Observe that the value of θ that maximizes $L(\theta)$ also maximizes $\ln L(\theta)$.

Remark

Maximizing is limited to solving the equation:

$$\frac{d \ln L(\theta)}{d\theta} = 0$$

Strictly speaking this is a necessary condition, but not a sufficient condition.

Examples

9. Let $X \sim \text{alt}(p)$ with unknown parameter p .

Bernoulli-experiment

Since: $f(x; p) = p^x \cdot (1-p)^{1-x}$, for $x = 0, 1$

$$\begin{aligned} \text{it follows: } L(p) &= \prod_{i=1}^n f(x_i, p) \\ &= (p^{x_1} \cdot (1-p)^{1-x_1}) \cdot (p^{x_2} \cdot (1-p)^{1-x_2}) \cdot \dots \\ &\quad \dots \cdot (p^{x_n} \cdot (1-p)^{1-x_n}) \\ &= p^{x_1+x_2+\dots+x_n} \cdot (1-p)^{(1-x_1)+(1-x_2)+\dots+(1-x_n)} \\ &= p^{\sum_{i=1}^n x_i} \cdot (1-p)^{n-\sum_{i=1}^n x_i} \end{aligned}$$

So:

$$\ln L(p) = (\sum_{i=1}^n x_i) \cdot \ln p + (n - \sum_{i=1}^n x_i) \cdot \ln(1-p) \quad \Rightarrow$$

$$\frac{d \ln L(p)}{dp} = \frac{\sum x_i}{p} + \frac{n - \sum x_i}{1-p} \cdot -1 = 0 \quad \Leftrightarrow$$

$$p \cdot (n - \sum x_i) = (1-p) \cdot \sum x_i \quad \Leftrightarrow$$

$$\hat{p} = \frac{\sum x_i}{n}$$

Therefore, the **maximum likelihood estimator** of p is given by the [sample proportion](#):

$$\hat{p} = \frac{\sum_{i=1}^n X_i}{n}$$

10. Let $X \sim \exp(\lambda)$ with unknown parameter λ .

Since: $f(x) = \lambda \cdot e^{-\lambda x}$, for $x \geq 0$

it follows: $L(\lambda) = \prod_{i=1}^n \lambda \cdot e^{-\lambda x_i}$

$$= \lambda^n \cdot e^{-\lambda \sum x_i} \Rightarrow$$

$$\ln L(\lambda) = n \cdot \ln \lambda - \lambda \cdot \sum x_i \Rightarrow$$

$$\frac{d \ln L(\lambda)}{d\lambda} = \frac{n}{\lambda} - \sum x_i = 0 \Leftrightarrow$$

$$\hat{\lambda} = \frac{n}{\sum x_i}$$

$$= \frac{1}{\bar{x}}$$

Therefore, the **maximum likelihood estimator** of λ is given by the reciprocal of the [sample mean](#):

$$\hat{\Theta} = \frac{1}{\bar{X}}$$

11. Let $X \sim N(\mu; \sigma^2)$ with unknown parameter μ and known variance σ^2 .

Since: $f(x) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$, for $x \geq 0$

it follows: $L(\mu) = \frac{1}{(2\pi\sigma^2)^{n/2}} \cdot \exp\left\{-\frac{1}{2\sigma^2}(\sum(x_i - \mu)^2)\right\} \Rightarrow$

$$\ln L(\mu) = -\frac{n}{2} \cdot \ln(2\pi\sigma^2) - \frac{\sum(x_i - \mu)^2}{2\sigma^2} \Rightarrow$$

$$\frac{d \ln L(\mu)}{d\mu} = \frac{\sum(x_i - \mu)}{\sigma^2}$$

$$= \frac{(\sum x_i) - n \cdot \mu}{\sigma^2} = 0 \Leftrightarrow$$

$$\hat{\mu} = \frac{\sum x_i}{n} = \bar{x}$$

Therefore the **maximum likelihood estimator** of μ is given by the [sample mean](#):

$$\hat{\Theta} = \bar{X}$$

12. Let $X \sim N(\mu; \sigma^2)$ with unknown parameters μ and σ^2 .

$$\text{Since: } f(x) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}, \text{ for } x \geq 0$$

$$\text{it follows: } L(\mu; \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \cdot \exp\left\{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2\right\} \Rightarrow$$

$$\ln L(\mu; \sigma^2) = -\frac{n}{2} \cdot \ln(2\pi\sigma^2) - \frac{\sum (x_i - \mu)^2}{2\sigma^2} \Rightarrow$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = \frac{\sum (x_i - \mu)}{\sigma^2} = 0$$

and

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0$$

Therefore the **maximum likelihood estimators** of μ and σ^2 are given by:

$$\hat{\Theta}_{\mu} = \bar{X} \quad \text{and} \quad \hat{\Theta}_{\sigma^2} = \frac{1}{n} \sum (X_i - \bar{X})^2$$

Properties of a Maximum Likelihood Estimator

When the sample size n is large enough and if $\hat{\Theta}$ is the maximum likelihood estimator of the parameter θ , then:

- $\hat{\Theta}$ is an approximately **unbiased estimator** for θ .
- The **variance of $\hat{\Theta}$** is nearly as small as the variance of any other estimator.
- $\hat{\Theta}$ follows an approximately **normal distribution**.